Lecture 33: The Back End of the Fuel Cycle

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 Introduction: The "Green Goo" Myth

Popular culture (The Simpsons) depicts nuclear waste as glowing green liquid leaking from drums. This is scientifically incorrect for commercial fuel.

- **Commercial Waste:** Solid ceramic pellets (UO_2) clad in solid metal tubes (Zircaloy). It is chemically inert and cannot "leak" liquid.
- **Defense Waste:** Liquid sludge from weapons programs (Hanford/Savannah River). This *is* a gooey mess, but it is a legacy of the Cold War, not civilian nuclear power generation.

2 Quantifying Radioactivity (Review)

Before classifying waste, we must briefly review how we quantify the hazard. Recall from our earlier lectures on radioactive decay that the activity A of a sample is the rate of decay:

$$A = \lambda N = \frac{\ln(2)}{T_{1/2}} N$$

where N is the number of radioactive atoms, and $T_{1/2}$ is the half-life.

- **Becquerel (Bq):** The SI unit. 1 Bq = 1 decay per second. (A very small unit).
- **Curie (Ci):** The traditional unit, based on the activity of 1 gram of Radium-226.
  1 Ci = 3.7×10^{10} Bq.

Because Activity is inversely proportional to Half-life ($A \propto 1/T_{1/2}$), isotopes with very short half-lives are intensely radioactive (high Bq), whereas isotopes with million-year half-lives are relatively stable and emit very little radiation per second.

3 Classification of Waste

Not all radioactive trash is created equal. We classify based on hazard and origin.

3.1 Low-Level Waste (LLW)

- **Content:** Rags, filters, medical tubes, booties, tools, and reactor structural materials contaminated with trace isotopes or neutron activation products.

- **Stats:** Accounts for over 90% of the volume of all radioactive waste, but contains only $\approx 1\%$ of the total radioactivity.
- **Disposal:** Shallow land burial (e.g., Barnwell, SC or Clive, UT). It is essentially a highly monitored, secure landfill.

Key Neutron Activation Products in LLW

When stray neutrons strike the steel and concrete of the reactor vessel, stable atoms absorb them and become radioactive (Activation). These make up the bulk of structural LLW:

Isotope	Source Material	Half-Life	Primary Hazard
Cobalt-60 (^{60}Co)	Structural Steel / Valves	5.27 years	Intensely strong Gamma emitter
Iron-55 (^{55}Fe)	Reactor Pressure Vessel	2.73 years	X-ray emitter
Nickel-63 (^{63}Ni)	Stainless Steel components	100.1 years	Beta emitter
Carbon-14 (^{14}C)	Coolant / Moderator impurities	5,730 years	Beta emitter, biological uptake

Table 1: Common Activation Products found in Low-Level Waste (Reactor Components).

3.2 High-Level Waste (HLW) / Used Nuclear Fuel (UNF)

- **Content:** The highly radioactive spent fuel assemblies themselves.
- **Stats:** $< 1\%$ of the total volume, but contains 99% of the radioactivity.
- **Scale:** All commercial used fuel generated in the US since the 1950s ($\approx 90,000$ metric tons) would fit on a single football field stacked roughly 10 yards high.

4 The Physics of Decay: Composition of Used Fuel

Fresh PWR fuel is roughly 95% U-238 and 5% U-235. After 4-5 years in the reactor (at a standard burnup of $\approx 45-50$ GWd/MTU), the chemical composition has fundamentally changed. *(Note that the residence time is much longer than the refueling cycle of 18 - 24 months due to core shuffling: partially spent fuel is moved to different parts of the core for neutron flux balancing and shielding).* The engineering challenge evolves over time because different isotopes dominate the hazard at different timescales.

4.1 Fission Products: The "Intermediate" Sweet Spot of Danger

When a uranium atom splits, the resulting fragments are highly unstable and undergo rapid beta decay down their respective decay chains. The engineering hazard of these fission products is dictated by their half-lives.

The isotopes that cause the most immediate trouble are those with an **"intermediate"** half-life.

- **The Intermediate Hazard (High Heat & Activity):** Isotopes with very short half-lives (days or months) burn themselves out quickly while the fuel is still in the reactor. However, isotopes with intermediate half-lives of roughly 30 years (like ^{137}Cs and ^{90}Sr) sit in a dangerous "sweet spot." They live long enough that they don't go away quickly, but decay fast enough to have an intensely high radioactive activity. They are the primary source of the massive decay heat that requires the fuel to be actively cooled to prevent the Zircaloy cladding from catching fire and the fission products from getting out.

- **The "300 Year" Rule:** After roughly 10 half-lives (300 years), the activity of these dominant 30-year isotopes drops by a factor of $2^{10} \approx 1000$. The waste becomes significantly cooler and less intensely radioactive.
- **Long-Lived Fission Products (LLFPs):** While intermediate isotopes dominate the heat profile, a few specific fission decay chains end at isotopes with much longer half-lives (e.g., ^{99}Tc and ^{129}I). Because their half-lives are so massive, their actual rate of decay (activity) and heat generation are virtually zero. Their hazard is not heat, but rather their long-term environmental mobility if water enters a repository over geologic timeframes.

Key Fission Product	**Half-Life**	**Primary Hazard / Characteristics**
	Intermediate-Lived (The Heat and Acute Radiation Hazard)	
Cesium-137 (^{137}Cs)	30.1 years	Major gamma & heat source; mimics K in the body.
Strontium-90 (^{90}Sr)	28.8 years	Major beta & heat source; mimics Ca (bone seeker).
	Long-Lived (The Geological / Environmental Hazard)	
Technetium-99 (^{99}Tc)	211,000 years	Low activity, but highly mobile in groundwater.
Iodine-129 (^{129}I)	15.7 million years	Low activity, but biologically active (thyroid).

Table 2: Key Fission Products in end-of-life PWR fuel, split by their dominant hazard profiles.

Course Connection: The Technetium Paradox (Medicine vs. Waste)

Earlier in the course, we discussed **Technetium-99m** (^{99m}Tc) as the workhorse of nuclear medicine. How does the exact same isotope save millions of lives in hospitals but cause billion-dollar engineering headaches for nuclear waste repositories?

1. The Medical Miracle (^{99m}Tc): In imaging, patients are injected with the metastable state of the atom. It has a half-life of just **6 hours** and decays by emitting a clean 140 keV gamma ray (perfect for escaping the body to hit a camera detector). Because the half-life is so short, the specific activity is enormous, meaning doctors only need to inject a microscopically tiny mass of ^{99m}Tc to get a bright scan.

2. The Patient Safety Factor (^{99}Tc): When ^{99m}Tc emits its gamma ray, it drops to its ground state: standard **Technetium-99** (^{99}Tc), which is a beta-emitter. Why doesn't this beta radiation harm the patient?

- The half-life suddenly jumps from 6 hours to **211,000 years**.
- Because Activity is inversely proportional to half-life ($A \propto 1/T_{1/2}$), the actual rate of radioactive decay drops off a mathematical cliff. The tiny mass of ^{99}Tc left in the patient's body is decaying so unbelievably slowly that it is essentially stable on a human biological timescale (and the body flushes it out via the kidneys in a few days anyway).

3. The Repository Problem: While a hospital patient only processes a few nanograms of Technetium, a commercial nuclear reactor creates **macroscopic quantities (kilograms)** of ^{99}Tc as a direct fission product. For the engineers designing Deep Geologic Repositories (like Yucca Mountain or Onkalo) or cleaning up the liquid tank sludge at Hanford, ^{99}Tc is one of the most dangerous isotopes on the board. Because it lives for 211,000 years, and because it dissolves incredibly easily in water (forming the highly mobile pertechnetate ion, TcO_4^-), it is one of the most likely isotopes to "escape" a repository and migrate into the groundwater over geologic timeframes.

4.2 The "Long" Life: Actinides ($t > 10,000$ years)

Once the high-heat fission products decay away, what remains are the heavy elements created when U-238 absorbs a neutron without fissioning (Transuranic Actinides).

- **Hazard:** These are primarily alpha-emitters. Because of their long half-lives, they generate very little heat, but they remain highly **radiotoxic** if ingested into the body via contaminated water.

Actinide Isotope	Approx. Amount (% mass)	Half-Life
Uranium-238 (^{238}U)	$\approx 94.0\%$	4.47 billion years
Uranium-235 (^{235}U)	$\approx 0.8 - 1.0\%$	704 million years
Plutonium-239 (^{239}Pu)	$\approx 0.5 - 0.6\%$	24,100 years
Plutonium-240 (^{240}Pu)	$\approx 0.2 - 0.3\%$	6,560 years
Americium-241 (^{241}Am)	$\approx 0.05\%$	432 years
Neptunium-237 (^{237}Np)	$\approx 0.05\%$	2.14 million years

Table 3: Key Actinides in end-of-life PWR fuel (assumes $\approx 5\%$ initial enrichment).

5 Interim Storage Solutions

Because the US has no operating permanent repository, we rely on interim storage to handle the heat and radiation in the decades following discharge.

5.1 Step 1: Wet Storage (Spent Fuel Pools)

Freshly discharged fuel is intensely radioactive and generates massive amounts of decay heat (driven by the intermediate-lived fission products). It is far too thermally hot for passive air cooling.

- **The Setup:** Fuel is moved via underwater transfer canals into massive, 40-foot-deep pools inside the reactor or auxiliary building. The water is heavily borated to absorb neutrons and prevent accidental criticality in the tightly packed racks.
- **Active Cooling Requirement:** Unlike dry casks, spent fuel pools *require continuous active cooling.* Massive pumps circulate the pool water through heat exchangers to remove the megawatt-level decay heat of a freshly offloaded core.
- **Shielding:** The water provides excellent radiation shielding. The 20+ feet of water above the fuel stops essentially all gamma and neutron radiation; standing at the edge of the pool exposes a worker to less radiation than a commercial airline flight.
- **The Station Blackout (SBO) Risk:** Because the pools require active electrical pumps, a complete loss of power (SBO) is a severe threat. Without cooling, the decay heat will eventually heat the pool to boiling.
  - *The Fukushima Context:* During the 2011 Fukushima Daiichi disaster, the tsunami knocked out all backup generators. Operators lost both SFP cooling *and* the electrical instrumentation needed to read the water levels. There was immense global panic that the pool for Unit 4 (which contained a freshly offloaded, highly active core) was boiling dry and uncovering the fuel. If the Zircaloy cladding is exposed to steam/air at high temperatures, it will chemically combust (a zirconium fire), releasing massive amounts of highly mobile radioactive isotopes (like ^{137}Cs and ^{131}I) directly into the atmosphere.
- **Duration:** Fuel must remain in the pool for a minimum of 3–5 years, until the decay heat of the individual assemblies drops below the thermal limits for passive dry storage (typically < 20 kW per cask).

Technical Note: Advanced SMR Initial Storage

Alternative SMR designs (LMFR and HTGR) fundamentally alter the "Back End" entry point by utilizing intrinsic material properties to manage decay heat.

1. LMFR: In-Vessel Sodium Storage

Liquid Metal Fast Reactors (e.g., Natrium, PRISM) cannot use water for cooling due to the violent $Na + H_2O$ reaction.

- **Integrated Buffer:** Spent fuel is moved to **internal racks** at the periphery of the reactor vessel, submerged in the primary sodium pool.
- **Passive Heat Path:** Sodium's high boiling point ($\approx 880°$C) and high thermal conductivity allow decay heat to dissipate into the reactor's "cold pool" via natural convection. This utilizes the *Reactor Vessel Auxiliary Cooling System* (RVACS), providing a passive safety margin that eliminates the need for external, pump-dependent pools.

2. HTGR: The "Dry-to-Cask" Pipeline

High-Temperature Gas-Cooled Reactors (e.g., Xe-100) utilize TRISO pebbles which decouple storage from water-dependency.

- **Low Power Density:** A single TRISO containing pebble generates negligible decay heat compared to an LWR assembly. This allows the fuel to be air-cooled via passive radiation almost immediately upon discharge.
- **Ceramic Stability:** Because the SiC containment is stable up to $\approx 1,600°$C, the fuel does not require a "wet" environment to prevent cladding failure or oxidation.
- **Direct Vaulting:** Spent pebbles are discharged directly into **dry storage canisters**. These canisters are then filled with an inert gas, sealed, and placed in a shielded casement that allows cooling by natural air circulation, skipping the "Initial Wet Storage" step entirely.

5.2 Step 2: Dry Cask Storage (ISFSI)

Once the decay heat drops below a threshold (usually $<$ 20 kW/assembly), fuel is moved to an Independent Spent Fuel Storage Installation (ISFSI) on the power plant's property.

- **Design:** The current proposal/standard is a massive stainless steel canister welded completely shut and backfilled with inert helium. This inner canister is placed inside a massive concrete "overpack" (weighing upwards of 100 tons) for radiation shielding.
- **Cooling:** Passive natural convection. Cool outside air enters vents at the bottom of the concrete overpack, heats up against the steel canister, and exits the top. No pumps, no fans, no electricity required.
- **Robustness:** These casks are heavily over-engineered. They have been physical tested by Sandia National Labs to withstand full-speed locomotive crashes, jet impacts, and being engulfed in jet fuel fires without breaching.

6 Deep Geologic Repositories and Long-Term Storage

The ultimate scientific consensus for the long-term isolation of the heavy actinides ($> 10,000$ years) is a Deep Geologic Repository (DGR). The goal is to bury the waste deep in stable bedrock, far below the water table, so it decays to the background radioactivity level of natural uranium ore before it can ever migrate to the surface.

6.1 The U.S. Saga: Yucca Mountain

In 1987, Congress amended the Nuclear Waste Policy Act to designate Yucca Mountain, Nevada, as the sole permanent repository for all U.S. commercial and defense high-level waste.

- **The Science:** Located in a remote desert ridge, the proposed repository was in welded tuff rock, sitting hundreds of feet *above* the deep water table, making it incredibly dry.
- **The Politics:** The project faced intense local opposition (coined the "Screw Nevada Bill"). Despite billions of dollars spent on tunnel boring and scientific validation proving its safety, the project was functionally defunded and abandoned in 2010 due to political pressure from the Nevada congressional delegation.
- **Current Status:** The US currently has no legal path forward for a permanent repository. Several private companies are currently attempting to license "Consolidated Interim Storage Facilities" (CISF) in Texas and New Mexico to centralize the dry casks, but these are tied up in intense state-level legal battles.

6.2 The Success Story: Finland's Onkalo

While the U.S. stalled, Finland succeeded. The **Onkalo** repository (located near the Olkiluoto nuclear plant) is the world's first operational deep geologic repository for commercial spent fuel.

- **The Method (KBS-3):** Fuel assemblies are placed into cast-iron inserts, which are then sealed inside massive, pure **copper** canisters (highly resistant to corrosion in oxygen-depleted environments).
- **The Isolation:** The copper canisters are lowered into holes drilled in 1.9-billion-year-old solid granite bedrock (over 400 meters underground) and packed tightly with bentonite clay, which swells when wet to perfectly seal out groundwater.
- **Timeline:** Operations are expected to commence in the mid-2020s, proving that the DGR concept is politically and technically achievable when local communities are treated as consenting partners rather than targets of federal mandates.

6.3 Nature's Proof: The Oklo Natural Reactor

When the public expresses doubt that we can contain radioactive isotopes for thousands of years, nuclear engineers point to Gabon, Africa.

- **The Natural Reactor:** Roughly 2 billion years ago, the natural abundance of U-235 on Earth was about 3% (comparable to modern PWR fuel). In the Oklo region of Gabon, groundwater seeped into a rich uranium ore deposit, acting as a moderator and initiating a spontaneous, natural nuclear chain reaction.

- **The Run Time:** This natural reactor pulsed on and off for hundreds of thousands of years, generating tons of highly radioactive fission products and transuranic actinides (like Plutonium).
- **The Result:** Modern geological surveys of the site reveal a stunning fact: over the last 2 billion years, without any engineered copper canisters or concrete overpacks, the highly dangerous plutonium and heavy actinides migrated **less than 10 meters** from where they were created. The natural surrounding rock safely contained the "waste" until it decayed entirely away.

7 Closing the Loop: Reprocessing

In the US, we use a "Once-Through" cycle (Mine → Burn → Bury). France, Russia, and Japan "close" the cycle.

7.1 PUREX (Plutonium Uranium Redox EXtraction)

- Fuel is dissolved in nitric acid.
- Organic solvents (TBP: Tri-Butyl Phosphate diluted in kerosene) chemically separate Uranium and Plutonium from the fission products which remain in the aqueous phase.
- **Pros:** Reduces the long-term radiotoxicity of the waste (by removing the Plutonium); Recovers massive amounts of energy (U and Pu can be recycled as MOX fuel).
- **Cons:** Proliferation risk (separates pure, weapons-usable Plutonium); Creates large volumes of highly acidic liquid chemical waste that must be managed.

8 The Legacy Challenge: Hanford Site

While commercial waste is solid, the US government has a massive inventory of liquid **Defense Waste** from the Cold War.

8.1 The Problem

- **Location:** Hanford, WA (Plutonium production for the Manhattan Project and Cold War).
- **Inventory:** 56 million gallons of chemical/radioactive sludge in 177 aging underground tanks.
- **Leakage:** Many single-shell tanks have leaked, threatening the Columbia River aquifer.

8.2 The Engineering Failure: The Pretreatment Facility (PT)

The original cleanup plan required a massive facility to separate the highly radioactive sludge from the liquid supernate using **Pulse Jet Mixers (PJMs)**. Construction began in 2002, but the facility is currently stalled (and may never operate) due to unresolved challenges in multiphase fluid mechanics and criticality safety.

Case Study: Stratification and the "Inhale" Criticality Risk

The PJMs act like massive mechanical lungs, cyclically drawing waste into a tube and then forcefully expelling it to mix the tank contents. However, engineers identified a fundamental challenge associated with the behavior of heavy actinide solids:

> **The Physics Problem:** Plutonium oxide particles are effectively very dense relative to the surrounding slurry ($\rho \approx 11.5$ g/cm^3 for solid PuO_2), leading to strong gravitational settling tendencies.
>
> - **Stratification:** Due to scaling limitations (captured by a high Richardson number, where buoyancy forces dominate over inertial mixing), the PJMs could not reliably maintain full suspension of heavy solids. As a result, actinide-rich material tended to accumulate near the bottom of the tank, forming a stratified layer with elevated fissile concentration.
> - **The "Inhale" Risk:** During the suction stroke, the PJM could entrain this locally concentrated layer into the mixer tube, effectively pulling a higher-than-average concentration of fissile material into a confined region.
>
> **The Consequence:** The combination of increased fissile concentration, unfavorable geometry (a confined tubular region), and the presence of water (an efficient neutron moderator) creates a credible risk scenario in which the system could approach criticality locally.
>
> Because nuclear criticality safety requires conservative guarantees under all operating conditions, the inability to rule out such localized configurations represented a major design barrier.

8.3 The Solution: Direct Feed (DFLAW)

After decades of difficulty in designing a system capable of safely processing the full waste stream, the DOE adopted a risk-reduction strategy focused on the low-activity fraction:

1. **DFLAW (Direct Feed Low-Activity Waste):**
   - Bypasses full sludge pretreatment. Only the liquid supernate is withdrawn from the top of the tanks.
   - Radioactive cesium is removed via ion exchange, and the resulting low-activity stream is sent directly to vitrification.
2. **Vitrification:**
   - The liquid waste is combined with glass formers (silica and boron compounds).
   - Melters operate at $\approx 1150°$C to produce stainless steel canisters filled with molten glass.
   - **Why Glass?** The radionuclides are incorporated directly into the amorphous glass structure. Even if the steel canister degrades over time, the glass matrix remains highly resistant to leaching and provides long-term structural stability on geologic timescales.

Engineering Contrast: The Hanford Legacy vs. Modern Reprocessing

A common misconception is that the intractable, leaking sludge tanks at Hanford are an unavoidable consequence of nuclear reprocessing. From a chemical engineering perspective, the "Hanford problem" is entirely a legacy of wartime material constraints.

The Hanford Mistake (Carbon Steel & Neutralization): During the Manhattan Project and the Cold War, the U.S. rushed to build massive underground waste tanks using cheap, readily available **carbon steel**. Because the PUREX waste stream is highly acidic (nitric acid), it would have rapidly eaten through the carbon steel. To prevent tank failure, engineers were forced to chemically *neutralize* the liquid by dumping in massive amounts of sodium hydroxide. This neutralization caused the heavy radioactive isotopes to precipitate out of the solution. The result was a heterogeneous, multi-layered nightmare of rock-hard saltcakes, thick sludges, and caustic liquids that is incredibly difficult to pump out and vitrify today.

Modern PUREX (Stainless Steel & Homogeneity): Modern commercial reprocessing facilities (e.g., La Hague in France, Rokkasho in Japan) do not neutralize their waste. Instead, they use actively cooled, double-walled **stainless steel** tanks, which are highly resistant to nitric acid corrosion. Because no neutralizing chemicals are added, the high-level waste remains a pure, homogeneous acidic liquid. This single-phase liquid is easily agitated, pumped, and fed directly into the high-frequency melters for vitrification into borosilicate glass.

Takeaway: The liquid waste from modern PUREX is a chemically simple fluid awaiting vitrification, not a solidifying sludge.

9 Decommissioning: What about the Reactor?

Students often ask: "Is the reactor vessel itself nuclear waste?" **Yes, but it is technically Low-Level Waste (LLW).**

9.1 The Distinction

- **High-Level Waste (HLW):** The Fuel Assembly (Pellets + Cladding + Grid structures). This is removed and stored in dry casks.
- **Low-Level Waste (LLW):** The piping, pumps, concrete, and the steel reactor vessel.

9.2 Categories of Hardware Waste

Depending on how close the metal was to the core (neutron flux), it falls into different buckets:

1. **Class A (95% of waste):** Concrete from the containment dome, tools, secondary piping. Low radioactivity. Buried in shallow trenches.
2. **The Reactor Vessel (Class B/C):** The massive steel pot is radioactive due to **Activation** (neutrons hitting iron/nickel atoms in the steel, see Table 1).
   - *Example:* The **Trojan Nuclear Plant** vessel was filled with concrete (to lock internal parts in place), welded shut, put on a barge, and buried whole in a dedicated trench at Hanford.

3. **Greater-Than-Class-C (GTCC):** The "Core Internals" (baffle bolts, core barrel) that sat inches from the fuel and sustained massive neutron bombardment.
   - These are so radioactive they cannot be legally buried in shallow trenches.
   - While legally classified as LLW, they are practically treated like HLW. They are often cut up underwater using remote submersibles and stored in dry casks alongside the spent fuel on the ISFSI pad.

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